

AKAOUI CHIHAB

SENIOR DATA SCIENTIST & ANALYTICS ENGINEER

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ABOUT ME :

Data Scientist and Data Analyst with strong experience in **marketing, CRM, e-commerce** and data engineering. I design end-to-end data solutions using **Python, SQL, cloud platforms, Power BI and ML models** to support customer knowledge, business performance and **strategic decisions**. My recent work focuses on **customer lifetime value**, appetite **scoring, marketing mix modeling**, causal analysis, promotional impact and **purchase journey detection**. I also work on **AI agents and LLM applications** to automate data and business processes. I am currently expanding my expertise in **MLOps and LLMs**. Outside work, I keep pushing my limits through trail running, boxing and continuous learning.

AREA OF EXPERTISE

- **Analytical Solutions:** Expertise in statistical modeling, behavioral analysis, predictive algorithms and machine learning applied to marketing, CRM and e-commerce use cases such as LTV, appetite scoring, segmentation, uplift and customer journey analysis.
- **Marketing Data Science:** Design and evaluation of models for customer lifetime value, promotional investment impact, marketing mix modeling, incremental revenue estimation, campaign performance and media channel contribution.
- **Data Engineering:** Designed and optimised robust ETL pipelines using dbt, Airflow and Spark to ensure high data quality. Strong experience in cloud data environments, including GCP and Azure, for reliable analytics and ML pipelines.
- **Dashboarding & Data Visualization:** Development of interactive and insightful dashboards using Power BI and Streamlit to support real-time decision-making.
- **Digital Marketing & CRM:** Optimised customer targeting, CRM campaigns, retention and engagement using data-driven insights, web analytics and campaign performance tracking.
- **Causal & Business Modeling:** Experience with causal approaches, including graph-based causal modeling, to better estimate commercial and promotional impact and support investment decisions.
- **Machine Learning & LLMs:** Experienced in applying traditional ML and Large Language Models (fine-tuning, RAG, prompt engineering...) using technologies such as Hugging Face, OpenAI, and Microsoft Azure AI Search for cloud-based AI solutions

INDUSTRY

- **Retail & E-commerce:** Customer knowledge, purchase journey analysis, promotional impact, CRM campaigns, media performance, LTV, segmentation and scoring use cases.
- **Banking & Insurance:** Experience with credit risk, CRM, customer analytics, SAS migration, process automation, lineage reconstruction and AI-assisted migration.
- **Energy & Mobility:** Fleet management, connected vehicles, charging stations, ecological transition.

TECHNICAL ENVIRONMENT

- **Languages :** Python, SQL, SAS
- **Orchestration & Processing:** dbt, Airflow, Databricks, Azure Data Factory, Spark
- **Cloud & Infrastructure:** GCP (BigQuery, Vertex, Dataflow,..), Azure (Synapse Analytics, AI Foundry, Logic App, Azure Bus...), AWS (S3, SQS), Snowflake, Docker
- **Versioning & Collaboration:** Git, Github Action, Jira, Office
- **Visualization:** Power BI, Streamlit, MicroStrategy
- **APIs & Integration:** API development, Postman, Azure API Management
- **Methodologies:** ETL, Data Modeling, CI/CD, RFI/RFP, Agile
- **Marketing & CRM:** Adobe Campaign, Adobe Analytics, Google Analytics, Odoo ERP

WORK EXPERIENCES

▸ DATA SCIENTIST - LEROY MERLIN

LILLE : 05/2025 - PRESENT

Project: Support the Customer Knowledge and Marketing team at Leroy Merlin in leveraging customer, CRM, media and e-commerce data through modeling, analysis and optimization. Focus on customer value, purchase behavior, marketing performance, promotional impact and data-driven decision-making, in collaboration with ADEO's data science teams.

Job Description

- **Data Science Referent:** Act as the main point of contact for data-related topics within the marketing team. Bridge between business stakeholders, data scientists, data engineers, product owners, and external partners.
- **Customer & E-commerce Analytics:** Analyze customer behavior, purchase journeys, CRM interactions and digital signals to identify business opportunities and improve customer knowledge.
- **LTV & Scoring Models:** Design and evaluate models for customer lifetime value, appetite scoring, segmentation, uplift and campaign response prediction.
- **Machine Learning & Recommendations:** Train and evaluate ML models for personalized recommendations, purchase journey detection and customer behavior prediction.
- **Marketing Mix Modeling:** Work on MMM approaches to estimate incremental revenue generated by media channels and support marketing investment decisions.
- **CRM & Campaign Performance:** Monitor CRM campaign performance, existing scores and segmentations such as RFM, CLTV, Uplift and MMM to support targeting and decision-making.
- **Promotion & Causal Modeling:** Modeled the impact of promotional and commercial investments using causal approaches, including graph-based causal modeling, to better estimate incremental revenue and support investment decisions.
- **Industrialisation & FinOps:** Support the industrialisation of models and monitor GCP cost impact, including Vertex AI, Compute Engine, BQ.
- **Visualisation & Sharing:** Build dashboards in Power BI and Data Studio to make results accessible to non-technical teams.
- **External Partner Oversight:** Review proposed methodologies, validate deliverables, and ensure industrialisation aligns with internal standards (robustness, reproducibility, cost-efficiency, scalability).

Technical environment: *Python, SQL, TensorFlow, Scikit-learn, GCP (BigQuery, Vertex AI, Compute Engine, GCS), Power BI, Power Query, Data Studio, Git, MLflow, dbt, Terraform*

▸ DATA ENGINEER | DATA SCIENTIST - ALLIANZ BENELUX

REMOTE : 10/2025 - 01/2026

Project: Contribute to the modernization of Allianz Benelux legacy SAS data flows by analyzing SAS programs, reconstructing lineage and supporting the migration of SAS inputs and outputs to a Snowflake architecture using dbt, Python and SQL. Bring automation and AI-assisted approaches to accelerate migration analysis and documentation.

Job Description

- **SAS Migration Assessment & Target Design:** Contributed to the migration approach for legacy SAS workflows, translating SAS programs, macros, inputs and outputs into Snowflake and dbt-compatible assets.
- **Legacy Code Audit & Lineage Analysis:** Analyzed SAS programs, lineage files and execution chains to identify dependencies, intermediate datasets, inputs, outputs and migration priorities.
- **Automation & AI Agents:** Explored AI-assisted approaches and AI agents using OpenAI to accelerate SAS code analysis, metadata extraction, dependency mapping and migration preparation.
- **SAS Expertise within the Team:** Acted as the main contributor on SAS-related topics, helping interpret DATA steps, PROC SQL, macros, includes, and legacy transformation logic.
- **Architecture & Migration Pattern Design:** Worked with senior data engineers on reusable migration patterns using Snowflake, dbt, Python and SQL.
- **Validation & Documentation Support:** Helped define validation principles to compare Snowflake/dbt outputs with SAS results, and produced technical notes to support the team.

Technical environment: *Python, SQL, SAS, dbt, Snowflake, Git, Azure Cloud, OpenAI*

▸ DATA ANALYST | ENGINEER - TOTALENERGIES

LILLE : 05/2023 - 01/2025

Project: Deliver advanced fleet management and connected vehicle solutions that drive TotalEnergies' ecological transition goals. Develop robust data pipelines, dashboards, and analytics to improve operational efficiency, support partner onboarding, and enable data-driven decision-making.

Job Description

- **Data Engineering & ETL:** Created and optimised ETL pipelines to process data from smart vehicles and charging stations. Used dbt for data transformation and Airflow to automate workflows, ensuring continuous quality control. Also leveraged Azure Synapse Analytics for data exploration and transformation, and contributed to implementing a Data Fabric architecture to ensure unified, governed, and accessible data across cloud platforms.
- **Dashboard & Analytics:** Designed real-time dashboards (Smart Vehicle, partner tracking, invoicing) with Streamlit and Power BI, providing immediate operational insights and supporting business decisions.
- **Business Value & Analysis:** Partnered with business teams to convert data insights into actionable strategies that reduce costs and enhance fleet performance.
- **Cross-Cloud Ingestion:** Facilitated consistent data flows between AWS (SQS) and Azure (Service Bus), ensuring scalable and reliable data ingestion across environments.
- **Algorithms & Modeling:** Built and industrialised models for travel, consumption, charging profiles, and range estimation, enhancing fraud detection and optimising fleet electrification.
- **RFI/RFP & Market Research:** Conducted targeted market analyses, prepared RFI/RFP documents, and guided strategic investments for partner integration (OEMs, TSPs).

- **API Management:** Developed and documented APIs to streamline data exchange, improving interoperability between diverse systems and partners.
- **Data Migration:** Led migration of partner data resources (OEMs, charging stations), upholding data integrity through automated monitoring and validation.

Technical environment: *Python, SQL, Databricks, Power BI, API, DBT, Azure Cloud (Synapse Analytics, Key Vault, Azure Data Factory, etc.), Power Query, Airflow, SQS, Git.*

▸ EXPERT DATA ANALYST - CREDIT AGRICOLE DU NORD

LILLE 07/2022 - 04/2023

Project: Deliver actionable analytics to align commercial and digital marketing strategies with customer needs, focusing on performance metrics and predictive modeling.

Job Description

- **Dashboards & Automation:** Created and automated performance dashboards (web traffic, commercial activity) to provide a coherent view of the marketing strategy.
- **Predictive Modeling:** Designed and deployed scores/models to address challenges such as customer interest and attrition.
- **Customer Analysis:** Studied customer base, relational intensity, leads, and satisfaction to refine targeting and retention approaches.
- **SAS & Datamart Migration:** Executed the migration of SAS processes and the datamart, implementing automated code to drive the transition to the new version.
- **Data Enrichment:** Contributed to managing, enriching, and ensuring the reliability of the customer database to meet evolving business requirements.

Technical environment: *SAS, SQL, Python, MicroStrategy, Excel*

▸ DATA ANALYST | SCIENTIST - BANQUE POPULAIRE DU NORD

LILLE : 10/2018 - 06/2022

Project: Analyze customer data, digital journeys, and commercial strategy to design ML algorithms and enhance CRM performance. Contribute to data community workshops and drive analytics adoption.

Job Description

- **ML Algorithms:** Developed and implemented predictive models (Scoring, Segmentation, NLP, Recommendation Engine).
- **Reporting & Workshops:** Produced reports for decision-makers and led technical training at Data/Marketing community events.
- **Digital Marketing:** Planned and executed digital campaigns using A/B testing, web analytics (clicks, ROI, leads) to acquire new customers and boost loyalty.
- **Data & Strategy:** Analyzed customer data, digital journeys, and commercial strategies to optimize targeting and improve conversion rates.
- **CRM Optimization:** Improved the effectiveness of solicitations, raising CRM campaign performance.
- **Cross-Functional Support:** Guided business units in defining needs (scope, target, content) to propose data-driven marketing solutions.

Technical environment: *SAS, SQL, Python, Adobe Campaign, Adobe Analytics, Power BI, Git, Excel*

▸ STATISTICIAN ENGINEER - COFIDIS

LILLE : 03/2018 - 09/2018

Project: Manage credit risk and acceptance in the Risk Partner Strategy and Management domain.

Job Description

- **Credit System:** Created a credit granting system balancing acceptance rates and risk exposure.
- **Risk Segmentation:** Deployed a segmentation approach to manage store partner risk.
- **Risk Differentiation:** Established metrics to distinguish between client and vendor risk.

Technical environment: SAS, SQL, R, Excel, Qlik

▸ DATA ONLINE RATER - APPEN

REMOTE : 01/2018 - 06/2019

Project: Improve and evaluate the relevance of Facebook ads.

Job Description

- **Ad Optimization:** Improved the audience targeting of Facebook ads.
- **Effectiveness Measurement:** Evaluated the effectiveness and relevance of Ads campaigns.
- **Quality Control:** Provided feedback loops for data-driven ad adjustments.

▸ SALES AGENT - MAKINOX SARL

NADOR : 05/2015 - 08/2016

Project: Support the company's operations by managing production, client relations, and supplier coordination.

Job Description

- **Production Management:** Oversaw manufacturing, maintenance, and logistics processes.
- **Client & Supplier Relations:** Handled end-to-end communication and negotiations.
- **Administration:** Managed documentation and administrative follow-ups.

Technical environment: Odoo, Excel

BACKGROUND / DIPLOMAS

2016 – 2018 : **Master's** in Statistical Engineering – Faculté des Sciences de Lille

2014 – 2015 : **Bachelor's** in Applied Mathematics – Faculté de Casablanca

2011 – 2013 : Preparatory Classes MP – Oujda

SIDE PROJECTS

- 2024-2025: Developed **AI agents** using LLM (OpenAI, Microsoft, Hugging Face, LangChain) and deployed them on **Azure** to automate tasks and improve user interaction.
- 2020-2023: Built a **crypto trading bot** (Python) featuring API integrations for real-time data and automated backtesting.
- 2018-2021: Built a dashboard for **cryptocurrency tax reporting**.
- Created **e-commerce** sites (Shopify, Etsy).

CERTIFICATIONS

- **2025** - **BigQuery** for Data Analysts (Google Cloud)
- **2025** - Multi **AI Agent** Systems with CrewAI (DeepLearning.AI)
- **2025** - Advanced **Data Engineering** (Coursera)
- **2025** - Operationalizing **LLMs on Azure** (Coursera)
- **2025** - Introduction to **Generative AI** (Coursera)
- **2024** - **Python** Essentials for **MLOps** (Coursera)
- **2024** - Machine Learning in Production (DeepLearning.AI)
- **2022** - **GCP** Modernizing **Data Lakes** and Data Warehouses (Coursera)
- **2022** - **GCP** Smart Analytics, **Machine Learning** and AI (Coursera)
- **2022** - Blockchain Scalability and its Foundations in Distributed Systems (Coursera)
- **2022** - Introduction to Blockchain for Financial Services (Coursera)
- **2022** - Bitcoin and Cryptocurrency Technologies (Coursera)
- **2021** - AWS Cloud Technical Essentials (Coursera)
- **2021** - Supply Chain Finance and Blockchain Technology (Coursera)
- **2021** - Python for Financial Analysis and Algorithmic Trading (Udemy)
- **2020** - Scala and Spark for Big Data and Machine Learning (Udemy)
- **2020** - Python for Data Science and Machine Learning Bootcamp (Udemy)
- **2019** - Machine Learning (Coursera)
- **2019** - Foundations of Marketing Analytics (Coursera)
- **2019** - Foundations of Strategic Business Analytics (Coursera)
- **2018** - Python Core (Sololearn)
- **2018** - SQL (Sololearn)

LANGUAGES

English : Fluent
Arabic : Bilingual

French : Bilingual
Spanish : Basic

ACTIVITIES

- Trail & Running, Boxing, Stock Market